

Total number of pages: 1
Degree: B.Tech

Roll no.:
Semester: 3

Mid Semester Examination, September 2024

Course Code: EIMTC301/ITMTC301/INMTC301
Course title: Optimization principles and techniques

Time: 1.5 hours

Maximum Marks: 20

Note: Missing data/information(if any), may be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	An electronic company is engaged in the production of two components C_1 and C_2 used in the TV sets. Each unit of C_1 costs the company Rs 25 in wages and Rs 25 in material, while each unit of C_2 costs Rs 125 in wages and Rs 75 in material. The selling price of C_1 is Rs 150 per unit and of C_2 is Rs 350 per unit. The company's production capacity is limited by two considerations. The company has initial balance as Rs 20,000. The company has 4000 hours of machine time and 2,800 hours of assembly time. The production of C_1 requires 6 hours of machine time and 4 hours of assembly time, whereas the production of C_2 requires 4 hours of machine time and 6 hours of assembly time. Formulate this problem as LPP so as to maximize total profit to the company.	2	CO
1b	Use graphical method to solve the LPP. $\text{Max } Z = 2x_1 + 4x_2$ subject to $x_1 + 2x_2 \leq 5$ $x_1 + x_2 \leq 4,$ $x_1, x_2 \geq 0.$	2	CO
2a	Use two-phase simplex method to maximize $\text{Max } Z = 5x_1 - 4x_2 + 3x_3$ subject to $2x_1 + x_2 - 6x_3 = 20$ $6x_1 + 5x_2 + 10x_3 \leq 76$ $8x_1 - 3x_2 + 6x_3 \leq 50$, $x_1, x_2, x_3 \geq 0.$	2	CO
2b	Find degenerate basic feasible solutions to the system $x_1 + x_2 + x_3 = 3$ $x_1 - x_2 + x_4 = 0$ $x_1, x_2, x_3, x_4 \geq 0.$	2	CO
3a	State Complimentary Slackness Theorem. Consider the following LPP: $\text{Max } Z = 3x_1 + 4x_2$ subject to $x_1 + x_2 \leq 12$ $2x_1 + 3x_2 \leq 30$ $x_1 + 4x_2 \leq 36, \quad x_1, x_2 \geq 0.$ The solution to the above LPP is $x_1 = 6, x_2 = 6$, and $Z = 42$. Write the dual of the above LPP. Also find the solution of the dual problem using complimentary slackness theorem.	2	CO
3b	Solve the following linear programming problem using dual simplex method. $\text{Min } Z = 4x_1 + 7x_2$ subject to $2x_1 + 3x_2 - x_3 = 5$ $x_1 + 7x_2 - x_4 = 9$ $x_1, x_2, x_3, x_4 \geq 0.$	2	CO

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4a	State the conditions in an optimal simplex table that infer: 1. Solution is infeasible. 2. Solution is unbounded. 3. Existence of alternate optima. 4. Solution is optimal	2	CO2																																				
4b	Find a feasible solution for the transportation problem given below using Vogel approximation method. <table><tr><td></td><td><i>D1</i></td><td><i>D2</i></td><td><i>D3</i></td><td><i>D4</i></td><td><i>Supply</i></td></tr><tr><td><i>O1</i></td><td>4</td><td>6</td><td>8</td><td>8</td><td>40</td></tr><tr><td><i>O2</i></td><td>6</td><td>8</td><td>6</td><td>7</td><td>60</td></tr><tr><td><i>O3</i></td><td>5</td><td>7</td><td>6</td><td>8</td><td>50</td></tr><tr><td><i>Demand</i></td><td>20</td><td>30</td><td>50</td><td>50</td><td></td></tr></table>		<i>D1</i>	<i>D2</i>	<i>D3</i>	<i>D4</i>	<i>Supply</i>	<i>O1</i>	4	6	8	8	40	<i>O2</i>	6	8	6	7	60	<i>O3</i>	5	7	6	8	50	<i>Demand</i>	20	30	50	50		2	CO2						
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5a	State degeneracy in the solutions to transportation problem. Use MODI method to find the optimum quantity to be supplied from each shop to different warehouses at minimum total cost for the following cost matrix, using initial cell allocations as $x_{11} = 20$, $x_{12} = 20$, $x_{22} = 10$, $x_{23} = 50$, $x_{34} = 50$. <table><tr><td></td><td><i>D1</i></td><td><i>D2</i></td><td><i>D3</i></td><td><i>D4</i></td><td><i>Supply</i></td></tr><tr><td><i>O1</i></td><td>4</td><td>6</td><td>8</td><td>8</td><td>40</td></tr><tr><td><i>O2</i></td><td>6</td><td>8</td><td>6</td><td>7</td><td>60</td></tr><tr><td><i>O3</i></td><td>5</td><td>7</td><td>6</td><td>8</td><td>50</td></tr><tr><td><i>Demand</i></td><td>20</td><td>30</td><td>50</td><td>50</td><td></td></tr></table>		<i>D1</i>	<i>D2</i>	<i>D3</i>	<i>D4</i>	<i>Supply</i>	<i>O1</i>	4	6	8	8	40	<i>O2</i>	6	8	6	7	60	<i>O3</i>	5	7	6	8	50	<i>Demand</i>	20	30	50	50		2	CO2						
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5b	We have to assign five jobs, I, II, III, IV, V to five workers A, B, C, D, and E. The time taken by different workers (in hours) in completing different jobs is given below: <table><tr><td></td><td><i>I</i></td><td><i>II</i></td><td><i>III</i></td><td><i>IV</i></td><td><i>V</i></td></tr><tr><td><i>A</i></td><td>160</td><td>130</td><td>175</td><td>190</td><td>200</td></tr><tr><td><i>B</i></td><td>135</td><td>120</td><td>130</td><td>160</td><td>175</td></tr><tr><td><i>C</i></td><td>140</td><td>110</td><td>155</td><td>170</td><td>185</td></tr><tr><td><i>D</i></td><td>50</td><td>50</td><td>80</td><td>80</td><td>110</td></tr><tr><td><i>E</i></td><td>55</td><td>35</td><td>70</td><td>80</td><td>105</td></tr></table> Find the minimum time (in hours) taken by the workers to complete all jobs.		<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>	<i>A</i>	160	130	175	190	200	<i>B</i>	135	120	130	160	175	<i>C</i>	140	110	155	170	185	<i>D</i>	50	50	80	80	110	<i>E</i>	55	35	70	80	105	2	CO3
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